Forecasting Non-Additivity Before the Experiment: A Pre-Experimental Triage for Single-Cell Perturbation Modeling

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Abstract

Deep models for single-cell perturbation response are frequently reported to be no better, in aggregate, than a simple additive or mean-based baseline. Kalygina et al. formalize this observation with Baseline Saturation (BS): the fraction of the reproducible, achievable performance range that a matched baseline already captures for a given perturbation. Their diagnosis is post hoc, computed only after the perturbation has been measured. We ask the complementary, experiment-facing question: can we predict, before running a perturbation, whether it will be saturated (an additive baseline suffices) or resistant (non-additive, worth the modeling and measurement cost)? We reproduce the BS metric and validate it against the source (per-metric agreement within 0.01 on Norman19). We then build a two-stage system: a Stage-1 triage model that predicts BS from purely pre-experimental, gene-intrinsic features (baseline expression, protein-protein interaction connectivity, Gene Ontology annotation, and a frozen protein language model embedding), and a Stage-2 router that sends predicted-saturated perturbations to the free baseline and predicted-resistant ones to a costlier model. On the genome-scale Replogle K562 essential screen (1,086 single-gene perturbations) the triage achieves Random Forest Spearman rho 0.65 (95% CI [0.61, 0.69]; the more conservative Ridge model reaches 0.58) and ROC-AUC 0.86 (label-permutation $p = 0.003$). For double perturbations, on the Wessels CaRPool-seq screen (158 doubles) a gene-pair triage predicts resistance at Ridge Spearman rho 0.50 (ROC-AUC 0.77, permutation $p = 0.003$). We find that saturation is largely a between-dataset and between-cell-type property, that which pre-experimental feature predicts resistance is itself context-dependent, and that, using a reconstructed reliability filter, the low cross-cell-type transfer of saturation appears to reflect genuine biology rather than measurement noise.

Keywords Baseline Saturation, Single-cell perturbation, Pre-experimental triage, Gene-pair prediction, CRISPR screening.

1. INTRODUCTION

Single-cell CRISPR screens with transcriptomic readout (Perturb-seq) have made it possible to train models that predict a perturbation's genome-wide expression response. A recurring and uncomfortable result is that these models, evaluated fairly, often fail to beat a simple baseline: for a double perturbation, the sum of the two single-perturbation effects (an additive baseline); for a held-out single perturbation, the mean over the other perturbations plus a target-gene scaling term. Ahlmann-Eltze, Huber and Anders [1] showed the additive baseline is not beaten in aggregate; Kalygina et al. [2] sharpened this into a per-perturbation diagnostic they call Baseline Saturation.

Baseline Saturation asks, for one perturbation and one evaluation metric, how much of the reproducible headroom the baseline already covers. The headroom is bounded below by a trivial "predict zero" model and above by a technical-duplicate positive control (one random half of the perturbation's cells predicting the other half), which captures

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the ceiling imposed by finite-cell sampling noise. A perturbation whose baseline reaches that ceiling is saturated: no model can do better on reproducible signal, so it is a poor benchmark and an unrewarding experiment to model. A perturbation whose baseline falls well short is resistant: there is real, reproducible, non-additive structure for a model to capture.

Kalygina et al. establish this diagnosis retrospectively. Our contribution is to make it prospective. If saturation could be predicted from information available before the perturbation is run, an experimentalist could triage a candidate list: spend deep-model compute and follow-up measurement on the resistant perturbations and fall back to the free baseline for the saturated ones. This reframes a benchmarking observation as a decision tool.

We contribute:

1. A validated open reimplementation of the Baseline Saturation metric and its controls (Section 3.1), reproducing the source per-metric values on Norman19 [3].
2. The finding that saturation is dominated by between-dataset and between-cell-type variation, so a useful triage must be trained across contexts (Section 4.1).
3. A Stage-1 triage model that predicts BS from pre-experimental gene-intrinsic features, with rigorous evaluation (permutation tests, bootstrap intervals, leave-one-group-out ablation) at genome scale (Sections 4.2 and 4.3).
4. A quantified, reliability-controlled analysis of how saturation transfers across cell types (Sections 4.4 and 4.5).
5. A Stage-2 routing result on the resistant Wessels doubles [4] with two findings: a genuine, correctly-signed routing lift under the correlation metric (DEG-weighted Pearson), and a clean negative result under the magnitude-sensitive metric (CCC) showing that the non-additivity defining Wessels resistance is idiosyncratic per double, not learnable from the two singles and not recovered even by a purpose-built context-rich architecture (GEARS) (Section 4.6).
6. A gene-pair triage that predicts double-perturbation resistance before the double is run, from only the two single profiles, on a genuinely resistant double dataset (Section 4.7).

2. DATA

The first four datasets are the GEARS-processed [5] AnnData files distributed on Harvard Dataverse, the standard inputs to deep perturbation benchmarks. The last two are scPerturb-harmonized h5ad [6], adapted to the same format by a conversion script.

Table 1 summarizes the six datasets used in this study.

Wessels23 is the one double-perturbation dataset here with a genuine resistant class and is the basis of the gene-pair result (Section 4.7). Sunshine23 is a high-MOI screen whose doubles are too sparse for stable per-double pseudobulk (only 19 scorable, all saturated); we use its 178 clean single perturbations, and it serves as a shallow-screen stress test for the reliability filter (Section 4.5). A second replication attempt, on the direct-capture replogle20 screen (GSE146194), was also examined but is excluded from the six datasets above and from the main results: it processed cleanly under the identical protocol yet proved near-fully saturated with no resistant class, so it has nothing to contribute to the triage and is reported only in Limitations (Section 6).

A hard limitation applies to absolute BS values off Norman: the source builds its gene panel as the union of the top approximately 8,192 highly variable genes and the target genes, computed from raw counts. The GEARS files expose only their own approximately 5,045 to 5,060 gene panel, and the full gene space has already been subset, so we cannot rebuild the 8,192-gene

Table 1. Datasets used in this study. "Regime" refers to the generalization setting defined by Kalygina et al. [2]: UnseenCombo holds out double perturbations whose constituent singles remain in training; UnseenPert holds out entire single perturbations.

Dataset	Cell line	Perturbation type	Regime	Unit	Scored perts
Norman19 [3]	K562	CRISPRa	UnseenCombo	double	124 (of 131)
Adamson16 [7]	K562	CRISPRi (UPR)	UnseenPert	single	86
Replogle22 K562 [8]	K562	CRISPRi (essential)	UnseenPert	single	1,086
Replogle22 RPE1 [8]	RPE1	CRISPRi (essential)	UnseenPert	single	1,533
Wessels23 [4]	THP-1	Cas13 (CaRPool-seq)	UnseenCombo	double	158
Sunshine23 [9]	Calu-3	Cas9 (host factors)	UnseenPert	single	178

panel from these files. Our metric math is validated on Norman (Section 3.1); absolute per-metric values on the other datasets are shifted upward by the smaller panel. BS remains faithful in structure and is a valid triage target.

3. METHODS

3.1. Baseline Saturation (Validated Reimplementation)

For a perturbation p we take the pseudobulk response $y_p = \mu_p - \mu_{\text{ctrl}}$ (mean expression under p minus mean control expression). Cells of each perturbation and of the control are split at random into halves A and B (fixed seed). Ground truth is the half-A response; the negative control is the zero vector; the positive control ("technical duplicate") is the half-B response, which predicts half A up to finite-cell noise. An interpolation-duplicate control is also computed and the better of the two positive controls is used per metric.

The matched baseline depends on the regime. For UnseenCombo (Norman doubles) it is the additive baseline: the sum of the two single-perturbation effects. For UnseenPert (single perturbations held out) it is the mean-over-perturbations baseline (leave-one-out mean of the other single effects) plus a global target-scaling term fit by ordinary least squares.

For each of seven headline metrics m (weighted MSE, weighted Pearson, weighted CCC, fraction of correct direction, cosine rank, GSEA-up, GSEA-down), Baseline Saturation is defined as:

$$s_{p,m} = \text{clip}_{[0,1]}\left(\frac{\bar{m}(b,y) - \bar{m}(\mathbf{0},y)}{\bar{m}(\hat{y}_{\text{pos}},y) - \bar{m}(\mathbf{0},y) + \boldsymbol{\varepsilon}}\right) \quad (1)$$

where $\bar{m}$ orients each metric so higher is better, b is the matched baseline prediction, $\hat{y}_{\text{pos}}$ is the positive control, $\mathbf{0}$ is the zero-vector negative control, and $\boldsymbol{\varepsilon}$ is a small constant for numerical stability. Metric weighting uses differentially expressed genes (DEGs) computed on the second-half split with scanpy's `rank_genes_groups` (`t-test_overestim_var`, reference "rest"), with per-gene weights from the squared, min-max-scaled t-scores. We aggregate the seven metrics to one scalar per perturbation by the median. Tiers follow the source: resistant BS < 0.33, frontier $0.33 \leq$ BS < 0.66, saturated BS ≥ 0.66.

Validation. On Norman19 UnseenCombo our per-metric median Baseline Saturation of the additive baseline matches the source (ours / paper): MSE 0.94 / 0.93, Pearson 0.98 / 0.97, CCC 0.97 / 0.96, fraction-direction 1.0 / 1.0, cosine-rank 1.0 / 1.0, GSEA-up 0.96 / 0.95, GSEA-down 0.96 / 0.95. The metric machinery is correct (Figure 1).

3.2. Reliability Filter (Reconstruction)

The source scores only perturbations whose measured effect is reliable, so that a low BS reflects non-additivity rather than noise. We reconstruct this with two per-perturbation statistics on the same half-split: a signal-to-noise ratio SNR $= \|y_p\|/\text{se}_p$, with se_p the summed per-gene sampling standard error (per-gene variance over cell count, for perturbation and control),

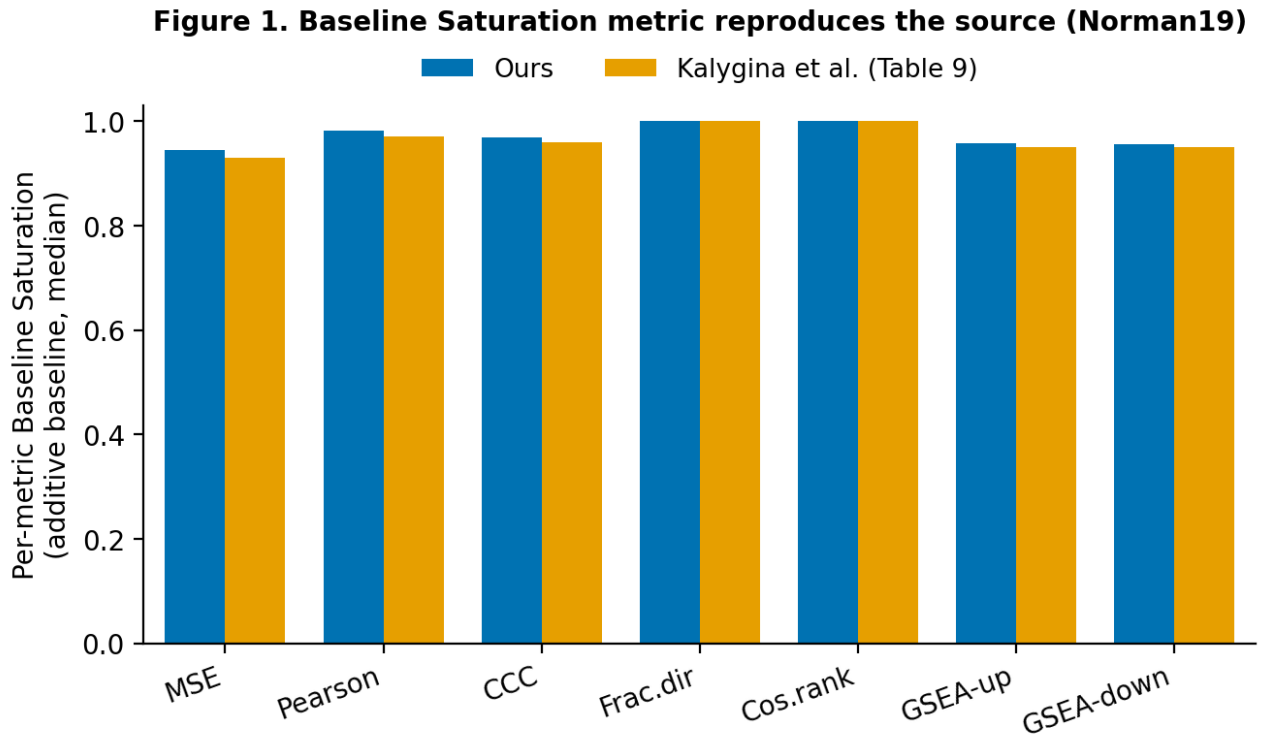


Figure 1. The Baseline Saturation reimplementation reproduces the source. Per-metric median Baseline Saturation of the additive baseline on Norman19 (UnseenCombo), our implementation (blue) versus the source (orange), across the seven headline metrics. Agreement is within 0.01 per metric. Our values are medians recomputed from the Norman saturation table. The source values are obtained by applying the Baseline Saturation formula (Equation 1) to the raw per-metric scores Kalygina et al. report for Norman19 in their Table 9 (the Zero, Tech-dup, and Additive rows; UnseenCombo, $n = 125$); the source does not tabulate Baseline Saturation per metric directly.

and a reproducibility ρ equal to the Pearson correlation between the half-A and half-B response vectors. Pass thresholds come from a control-only null: control cells are split into four random folds, two pseudo-effects are formed, and their SNR and ρ are recorded; the 95th percentile of each null is the threshold. A perturbation is reliable if it passes either criterion. Appendix H of the source gives the criteria but not all constants, so this is flagged as a reconstruction.

3.3. Pre-Experimental Features

The triage uses only features knowable before the perturbation is run. All are gene-intrinsic (properties of the perturbed gene, not its measured response).

- Baseline expression (from control cells only): mean, detection rate, variance of the target gene, and whether the target gene is itself measured in the panel.
- STRING protein-protein interaction [10]: degree, weighted degree, clustering coefficient, and mean edge confidence of the target protein.
- Gene Ontology: counts of biological-process, molecular-function and cellular-component annotations, and a transcription-factor flag (GO:0003700 / GO:0000981).
- Protein language model: mean-pooled frozen ESM-2 [11] (`esm2_t12_35M`, 480-dim) embedding of the target protein sequence. Used on Adamson only in this study.

For the gene-pair (double) framing we also compute pair features: the log-fold-change (LFC) relationship of the two single effects (cosine, dot product, norms), GO co-annotation Jaccard index, STRING shortest-path distance, and ESM cosine similarity.

3.4. Stage-1 Triage

We predict BS_{median} (regression) and the saturated-versus-not label (classification) from the features above. The classification tasks binarize BS_{median} at the same tier boundaries defined in Section 3.1 (resistant BS < 0.33, saturated BS ≥ 0.66), inherited directly from the source paper and not independently tuned for our feature set; a different cutpoint choice could shift the reported ROC-AUCs. Models are scikit-learn Ridge and RandomForest for regression and LogisticRegression for classification. The ESM block is reduced by PCA refit inside each cross-validation fold to prevent leakage. Cross-validation is leave-one-out for small n and shuffled 10-fold at genome scale; shuffling is essential because the feature tables are sorted by BS and unshuffled folds would be BS-contiguous and badly deflate the score. We report label-permutation p-values (300 shuffles), 1,000-sample bootstrap confidence intervals, and both cumulative and leave-one-group-out feature ablations.

3.5. Stage-2 Routing

The routing evaluation compares, on the complete test set and by DEG-weighted Pearson correlation to ground truth: baseline-only, Stage-2-only, routed-by-predicted-BS (predicted-saturated to the baseline, predicted-resistant to Stage-2), and an oracle router using true BS. For the single-perturbation setting the Stage-2 model is an ESM-similarity nearest-neighbour effect predictor. For the double-perturbation setting (Section 4.6) the Stage-2 model is a gradient-boosted regressor that predicts each gene's non-additive residual (the double effect minus the additive baseline) from the two single effects, so its output is the additive baseline plus a learned correction; doubles are held out by GroupKFold so a double's genes never span train and test. We additionally evaluate with DEG-weighted concordance correlation coefficient (CCC), which unlike Pearson is sensitive to magnitude errors. The framework is agnostic to the Stage-2 model.

3.6. Statistical Summary

Table 2 provides a consolidated view of every triage result reported in Section 4. Regression predicts BS_{median} by Spearman ρ under leave-one-out or shuffled 10-fold CV; classification is LogisticRegression binarized at the Section 3.1 tiers. Permutation tests use 300 label shuffles on the Ridge regressor; bootstrap CIs use 1,000 resamples. No single regressor is uniformly best across datasets (Section 6); we report both where both were run.

4. RESULTS AND DISCUSSION

4.1. Saturation Is Largely a Between-Dataset and Between-Cell-Type Property

Norman19 double perturbations are essentially fully saturated under every metric (the least-saturated pair still has weighted-Pearson BS approximately 0.88). The heterogeneity the source reports is a cross-dataset, cross-regime figure, not a within-dataset one: pooling the per-tier condition counts in its Table 1 across all nine datasets and five generalization regimes gives about 55% saturated, 24% frontier, and 21% resistant (6,025 / 2,655 / 2,313 of 10,993 held-out conditions). The additive baseline's weighted Pearson varies far more between datasets than within: Norman 0.960, Replogle20 0.996, Sunshine23 0.636, Wessels23 0.373. A triage model trained on any single saturated dataset would see almost no resistant class; both classes appear only across datasets, regimes and cell types (Figure 2). This motivates the multi-dataset design below. This is a statement about the base rate of saturation across contexts, not about its predictability: within a fixed dataset and cell type, saturation still carries learnable structure that pre-experimental features recover (Sections 4.2 and 4.3).

Table 2. Consolidated triage performance across all datasets. The classification task is saturated-versus-not for Adamson and both Replogle screens, but resistant-versus-not for Wessels (0.774; Wessels saturated-versus-not is 0.732). Permutation p-values are computed on the Ridge regressor; each Random Forest row cites that dataset's Ridge permutation, as no separate Random Forest permutation was run.

Dataset (unit)	n scored	Regressor	Spearman ρ [95% CI]	Perm p	ROC-AUC [95% CI]
Adamson (single)	86	Ridge	0.559 [0.38, 0.70]	0.003	0.739 [0.67, 0.86]
Replogle K562 (single)	1,086	Random Forest	0.650 [0.609, 0.688]	0.003 (Ridge)	0.863 [0.833, 0.892]
Replogle K562 (single)	1,086	Ridge	0.580 [0.534, 0.620]	0.003	as above
Replogle RPE1 (single)	1,533	Ridge = RF	0.353 [0.31, 0.40]	0.003	[0.66, 0.72]
Wessels gene-pair (double)	158	Ridge	0.502 [0.378, 0.609]	0.003	0.774
Wessels gene-pair (double)	158	Random Forest	0.510 [0.383, 0.620]	0.003 (Ridge)	as above

Figure 2. Saturation is largely a between-dataset / between-cell-type property

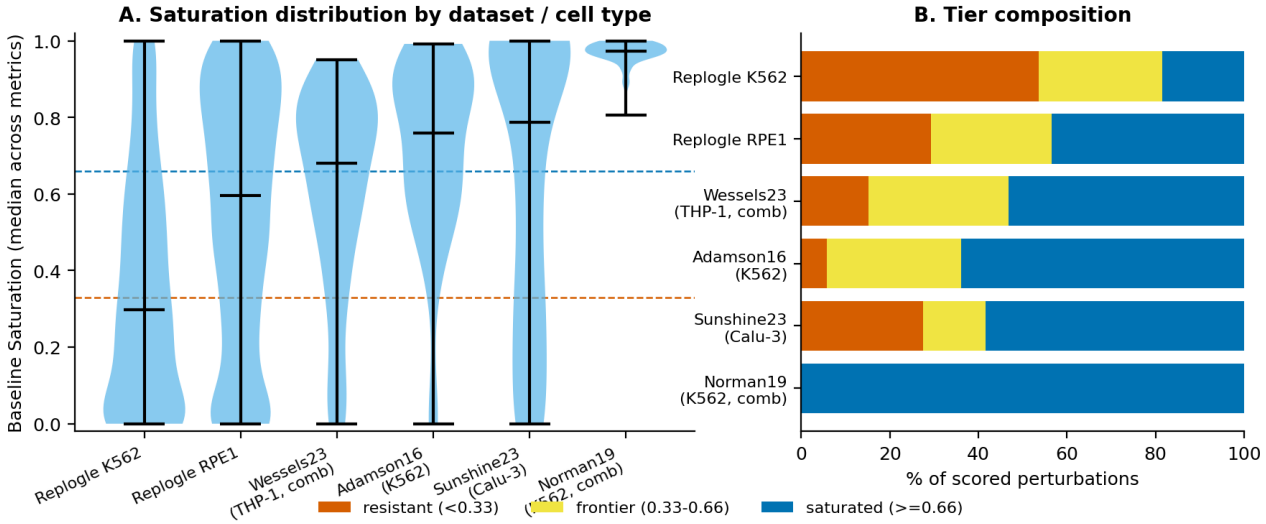


Figure 2. Saturation is largely a between-dataset / between-cell-type property. (A) Per-item Baseline Saturation distributions (median across metrics) for six screens spanning four cell lines and three assays, ordered by median; dashed lines mark the resistant (0.33) and saturated (0.66) tier boundaries. (B) Tier composition per screen. The resistant class is a substantial share of Replogle K562 and a smaller but real share of Sunshine, and is nearly absent in Norman. All per-item data from the saturation result tables.

4.2. Stage-1 Triage Predicts Baseline Saturation (Adamson, with Interpretable Biology)

Adamson16 is a UPR CRISPRi screen with 86 single-gene perturbations and both classes present (64% saturated / 30% frontier / 6% resistant). The most resistant perturbations are UPR transcription factors and effectors: DDIT3 (CHOP), CREB1, BHLHE40, ZNF326, HSPA5 (BiP), exactly the genes expected to drive idiosyncratic programs that a mean-over-perturbations baseline cannot reproduce.

The triage predicts $\mathrm{BS}_{\mathrm{median}}$ at Ridge Spearman $\rho = 0.559$ (leave-one-out CV), with a label-permutation p of 0.003 (null mean approximately 0) and a bootstrap 95% CI of [0.38, 0.70]. Classifying saturated versus not gives ROC-AUC 0.739, 95% CI [0.67, 0.86]. Cumulative ablation (ρ / AUC): expression 0.13 / 0.55, plus STRING 0.42 / 0.61, plus GO 0.53 / 0.68, plus ESM-2 0.56 / 0.74. Leave-one-group-out shows GO is the most important group here (drop -0.13, driven by the transcription-factor flag), then STRING (-0.06), expression (-0.03), ESM (-0.03). The picture is interpretable: hub, highly expressed genes are saturated (they drive shared programs the baseline already captures); transcription factors and peripheral genes are resistant.

4.3. The Triage Holds at Genome Scale (Replogle K562)

The Replogle K562 essential screen (162,751 cells, 1,092 single perturbations, 1,086 scored) is far less saturated than Adamson: median BS 0.30, split 19% saturated / 28% frontier / 54% resistant. The most resistant perturbations are core essential genes (mitochondrial NDUFAB1, NDUFB10, COX7C; transcription machinery GTF2A1/2), which is biologically sensible.

With gene-intrinsic features only (no ESM), the full rigorous suite gives: RandomForest Spearman 0.650, 95% CI [0.609, 0.688]; Ridge 0.580, 95% CI [0.534, 0.620]; classification ROC-AUC 95% CI [0.833, 0.892]; and a label-permutation p of 0.003 (observed 0.580 versus null mean -0.017, sd 0.041). This is far better powered than Adamson and confirms the thesis at genome scale.

The feature-group ablation reveals a context shift. On K562, expression dominates (leave-one-group-out drop -0.362), STRING adds a solid increment (drop -0.062), and GO is negligible (drop -0.005), the reverse of Adamson where the transcription-factor flag led. So not only saturation itself but which pre-experimental feature predicts it is context-dependent (Figure 3).

4.4. Saturation Transfers Only Partially Across Cell Types (Replogle RPE1)

The same essential screen in RPE1 (1,533 perturbations) is more saturated than in K562: median BS 0.60, split 44% saturated / 27% frontier / 29% resistant. For the 835 genes perturbed in both lines, label transfer is modest: Spearman(BS_{K562}, BS_{RPE1}) = 0.304 ($p = 3 \times 10^{-19}$), Pearson 0.318, with resistant-call agreement 58% (52% of shared genes resistant in K562 versus 18% in RPE1). Saturation is therefore partially gene-intrinsic but substantially cell-type-specific.

The predictive structure differs too. Within RPE1 the triage is much harder than in K562: Ridge and RandomForest both reach only $\rho \approx 0.353$, 95% CI [0.31, 0.40], with ROC-AUC approximately [0.66, 0.72] and permutation $p = 0.003$ (real but weaker). And the dominant feature group flips again: STRING/PPI connectivity leads in RPE1 (drop -0.116) where expression led in K562. Training the triage on one cell line and testing on the other drops performance sharply (K562 to RPE1 $\rho = 0.27$ / AUC 0.60; RPE1 to K562 $\rho = 0.32$ / AUC 0.69) relative to within-cell ($\rho = 0.64$ / AUC 0.86). The triage must be trained per context, or supplied with cell-type features.

4.5. The Cross-Cell-Type Gap Is Real, Not a Noise Floor (Reliability Filter)

A natural worry is that the low cross-cell-type transfer merely reflects noisy half-split BS estimates. Our reconstructed reliability filter (Section 3.2) lets us test this. On these deep screens the control noise floor is low (SNR $>$ approximately 1.05, $\rho >$ approximately 0.04 to 0.07) and the hard gate removes almost nothing (Adamson 100% reliable, both Replogle screens approximately 99.9% pass), so we use reliability as a graded quantity.

Two findings follow. First, reliability correlates positively with saturation: Spearman(reproducibility, BS_{median}) = $+0.35$ (K562), $+0.32$ (RPE1), and the resistant tier is enriched for low-reproducibility perturbations (64% of K562 and 76% of RPE1 resistant calls fall in the low-reproducibility half, versus 50% by chance). So a share of within-dataset resistant calls are noise-limited measurements, and the reported resistant fractions are upper bounds; a reliability-cleaned resistant set is smaller and more biological.

Second, and more importantly, the cross-cell-type label transfer is robust to reliability filtering. Restricting to genes reliable in both lines at rising reproducibility quantiles gives Spearman 0.304 (all 835 genes), 0.328 (top 75%, 626), 0.292 (top 50%, 418), 0.344 (top 25%, 209): robust and non-monotone, only marginally higher on the cleanest quarter. The low transfer is therefore not a measurement-noise floor; cell-type specificity of saturation appears to be a genuine biological finding, on the strength of

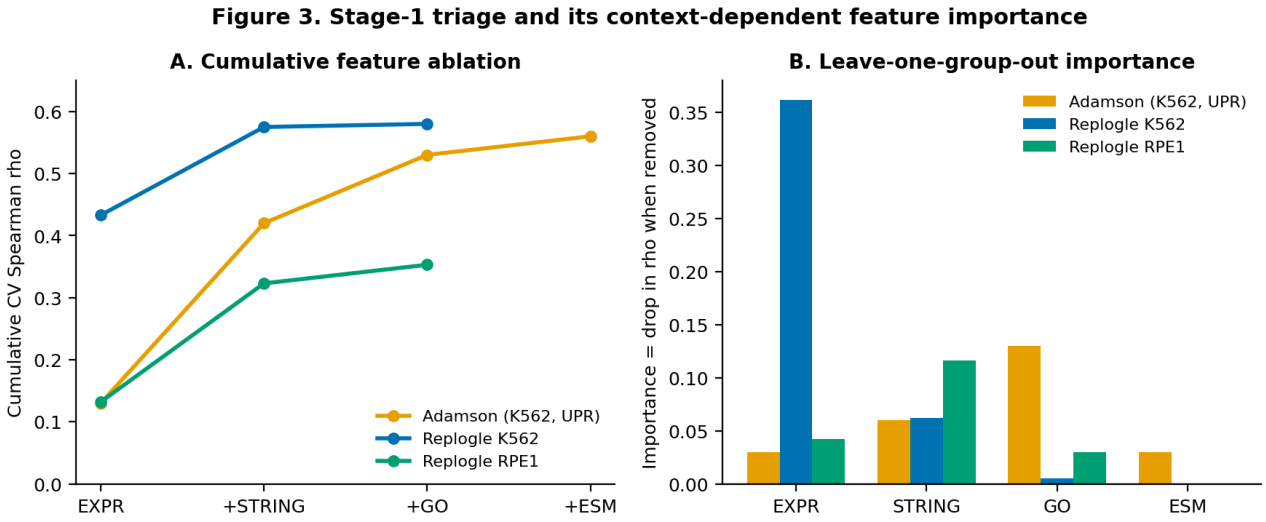


Figure 3. Stage-1 triage and its context-dependent feature importance. (A) Cumulative cross-validated Spearman ρ as feature groups are added (EXPR, +STRING, +GO, +ESM). (B) Leave-one-group-out importance (drop in ρ when a group is removed). The dominant predictor of resistance flips by context: expression in Replogle K562, PPI connectivity in Replogle RPE1, the transcription-factor flag (GO) in the Adamson UPR screen.

this reconstructed filter (Figure 4).

4.6. Stage-2 Routing: Framework and an Honest Null

On Adamson the routing comparison (DEG-weighted Pearson to ground truth) gives: baseline-only 0.752, Stage-2-only 0.639, routed 0.751, oracle-routed 0.753. The routing lift is approximately 0, which is the honest and expected outcome: a near-saturated dataset gives the baseline no headroom, and the simple ESM-nearest-neighbour Stage-2 does not beat the baseline even on the resistant subset, so even a perfect router cannot help. The routing framework is the reusable deliverable; a positive routing benefit requires (i) a dataset with a genuine resistant class with headroom and (ii) a Stage-2 model shown to beat the baseline on that subset.

On the resistant Wessels doubles we can run the routing experiment with headroom. Here Stage-2 is a trained gradient-boosted model that predicts the non-additive residual per gene from the two single effects, so its prediction is the additive baseline plus a learned correction (and falls back to additive when the interaction is not learnable). Two accuracy metrics tell a two-part story. Under DEG-weighted Pearson (scale-invariant, correlation structure), Stage-2 improves the resistant subset (+0.027) while staying neutral on the rest, lifting mean accuracy from 0.402 to 0.407, with oracle routing at 0.409 (+0.007 over additive): a small, real, correctly-signed benefit concentrated on the doubles the triage flags resistant. Under DEG-weighted CCC (which is sensitive to the magnitude overshoot that actually defines Wessels resistance), no learnable function of the two singles beats additive, and oracle routing cannot improve on it. The magnitude non-additivity is idiosyncratic per double, not a systematic interaction that generalizes across doubles (Figure 5B). This is itself the argument for triage: it identifies the doubles whose response cannot be recovered additively or by a simple trained model, i.e. the ones worth the experiment or a context-rich model.

We further tested whether a context-rich model closes the magnitude gap the residual model leaves open. We trained GEARS [5], a graph neural network over a GO perturbation graph and a gene co-expression graph, in the same leave-one-double-out setup (5-fold GroupKFold, both constituent singles always in training) and evaluated its predicted double effects under the identical DEG-weighted metrics. Because GEARS

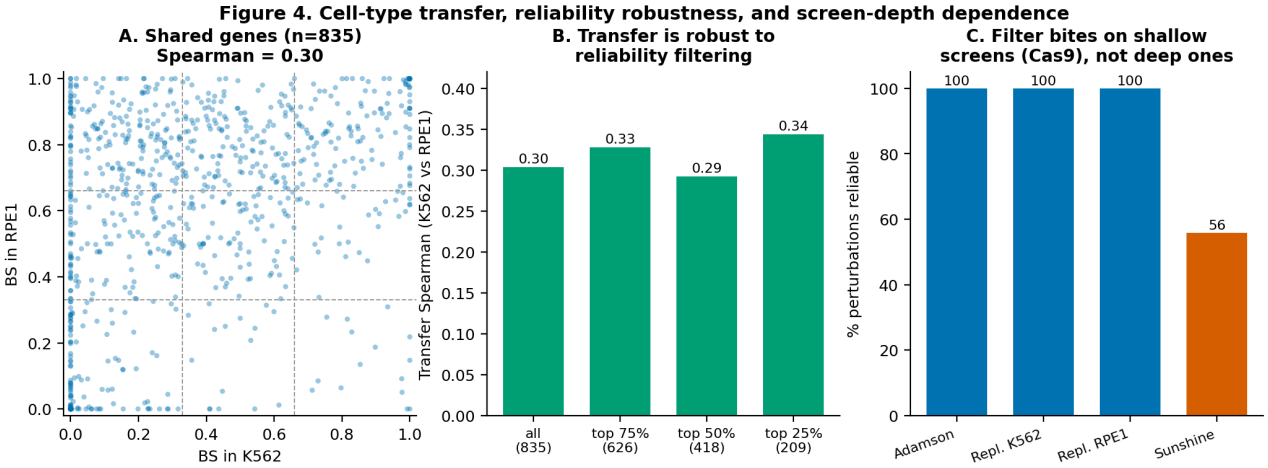


Figure 4. Cell-type transfer, reliability robustness, and screen-depth dependence. (A) Baseline Saturation of the 835 genes perturbed in both K562 and RPE1 (Spearman 0.30): partially gene-intrinsic, substantially cell-type-specific. (B) The K562-vs-RPE1 transfer correlation is robust to restricting to more reliable genes, so it is not a noise floor. (C) Fraction of perturbations passing the reliability filter, computed from the reliability results: the gate removes almost nothing on the deep screens but bites on the shallow Sunshine Cas9 screen.

restricts perturbations to GO-annotated genes, two of Wessels' 28 targets (CD14, VPRBP) are dropped, so this comparison runs on the 142 of 158 doubles whose two singles both survive that filter; on this subset the additive and residual-GBM baselines recompute to 0.411 and 0.416 under Pearson and 0.348 and 0.298 under CCC (and so are not directly comparable to the 158-double figures above). GEARS reaches 0.417 under Pearson, a +0.006 lift over additive that essentially matches the residual GBM and does not exceed it on the resistant subset, and 0.305 under CCC, which like the GBM stays below the additive baseline (−0.042). A purpose-built context-rich architecture therefore does not recover the magnitude non-additivity either: the negative result under CCC is robust to model class, not an artifact of the hand-engineered residual model.

4.7. Gene-Pair Triage: Predicting Double-Perturbation Resistance Before the Experiment

This is the double-perturbation counterpart of the Stage-1 triage and the target result of this study. Norman doubles are near-saturated, so we use Wessels CaRPool-seq (Cas13, THP-1), the one double dataset here with a genuine resistant class: 158 doubles with both constituent singles measured. Its UnseenCombo Baseline Saturation has pooled median 0.681 and tier split 53% saturated / 32% frontier / 15% resistant, and the magnitude-sensitive metrics are strongly resistant (median MSE-BS 0.139, weighted-Pearson-BS 0.377, CCC-BS 0.274), matching the source's reported additive weighted Pearson of approximately 0.37. The additive baseline gets direction, cosine and GSEA right (approximately 0.9) but magnitude wrong, the signature of non-additivity.

Using only pre-experimental gene-pair features (the log-fold-change relationship of the two already-measured single effects, GO co-annotation, and STRING PPI), the gene-pair triage predicts the double's Baseline Saturation at Ridge Spearman 0.502 (95% CI [0.378, 0.609]) and RandomForest 0.510 ([0.383, 0.620]) under leave-one-out CV, with a label-permutation p of 0.003 (observed 0.502 versus null mean −0.033). Classifying the operationally relevant call, resistant (BS < 0.33) versus not, gives ROC-AUC 0.774. Leave-one-group-out ablation shows the log-fold-change relationship of the two singles carries essentially all the signal (drop −0.269), while GO and STRING are negligible; the LFC block alone reaches $\rho = 0.522$ (Figure 5A). So an experimentalist can

predict, before running a double perturbation, whether it will be resistant (worth a complex model or the experiment) or saturated (use the free additive baseline), from information already in hand.

One caution against single-feature intuition: `lfc_norm_sum` correlates $+0.53$ with Baseline Saturation on Wessels (large-effect pairs more saturated) but -0.28 on Norman, so the sign of the LFC-magnitude relationship is itself regime-dependent. The multivariate model, not any one feature, is the deliverable.

5. DISCUSSION

Baseline Saturation turns "deep models do not beat baselines" from a discouraging aggregate into a per-perturbation property with structure worth predicting. Our results argue that this property is mostly not intrinsic to a gene: it moves strongly between datasets, regimes and cell types, and even the features that forecast it change with context. That is itself the actionable message for an experimentalist. Rather than asking "is this gene saturated," one should ask "is this gene saturated in this cell type and assay," and a cheap gene-intrinsic triage answers that well within a context (genome-scale Random Forest $\rho = 0.65$, classification AUC 0.86 in K562) while transferring only partially across contexts.

The practical use is allocation. Given a candidate list and a fixed budget, the triage ranks perturbations by predicted resistance, so scarce deep-model compute and follow-up measurement go where non-additive structure actually exists. Because the features are gene-intrinsic and cheap (public PPI, GO, a control-only expression profile, and a frozen protein embedding), the triage adds negligible cost.

The gene-pair triage extends this logic to combinatorial perturbation screens. From only the two already-measured single-perturbation profiles, we predict whether a double perturbation will be resistant or saturated (Spearman $\rho = 0.50$, resistant-versus-not AUC 0.77 on Wessels). The LFC relationship of the two singles carries essentially all the signal, so the practical requirement is minimal: measure the singles, then decide which doubles are worth running. The Stage-2 routing experiment on Wessels provides both a positive result (small, correctly-signed lift under Pearson concentrated on the triage-flagged resistant subset) and an honest negative result (no learnable function of the two singles recovers the magnitude non-additivity under CCC). This negative finding is itself informative: it sharpens the argument for triage, because it identifies the doubles whose biology genuinely cannot be recovered from cheaper surrogates. This negative result survives a strong test: even GEARS, a context-rich graph neural network built for exactly this prediction task, fails to recover the magnitude non-additivity on these doubles (Section 4.6), so it is the idiosyncrasy of resistant doubles, not a weakness peculiar to the gradient-boosted residual model, that drives the value of triage. When a resistant double's response cannot be predicted from its singles by either a simple trained model or a purpose-built deep architecture, flagging that double for the full experiment, rather than trusting any cheap surrogate, is the substantive recommendation.

The reliability filter analysis adds a methodological contribution, with the caveat that the filter itself is a reconstruction (Section 3.2). The low cross-cell-type transfer of saturation ($\rho \approx 0.30$ between K562 and RPE1) survives reliability filtering across the full quantile range (Spearman 0.304 to 0.344 as genes are restricted to more reliable subsets). Cell-type specificity of saturation therefore appears to be a genuine biological property rather than a measurement-noise floor, which strengthens the practical conclusion that the triage must be trained per experimental context, though this should be read as preliminary evidence given the reconstructed nature of the filter.

6. LIMITATIONS

Gene panel. We are locked to the GEARS approximately 5,045-gene panel and cannot

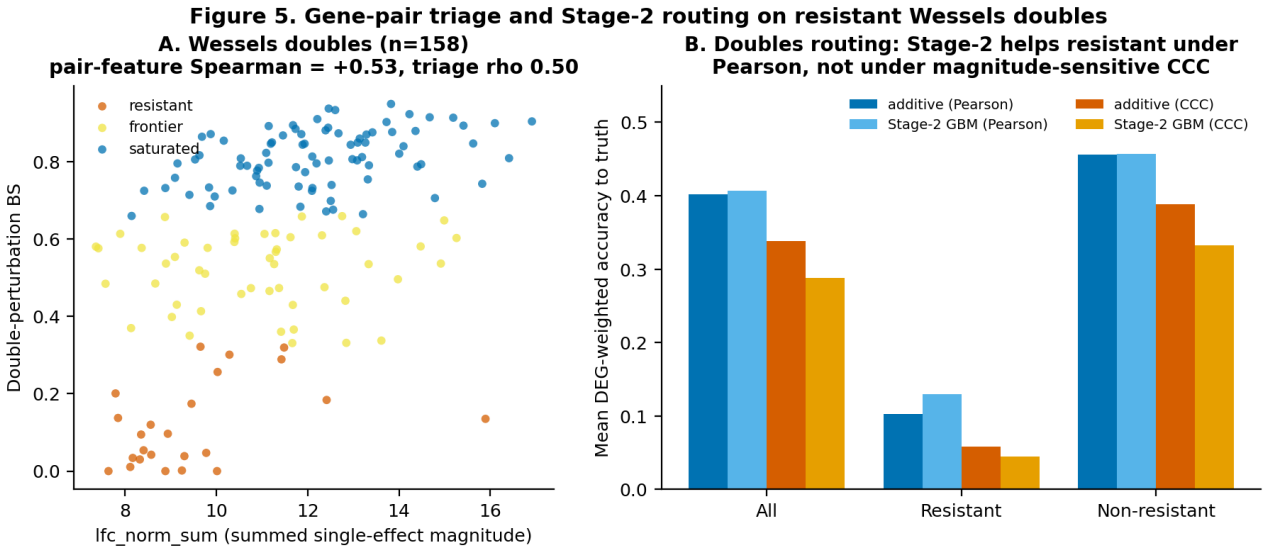


Figure 5. Gene-pair triage and Stage-2 routing on resistant Wessels doubles. (A) A single pair feature (summed single-effect magnitude) versus double-perturbation Baseline Saturation, colored by tier; the multivariate gene-pair triage reaches Spearman 0.50. (B) Doubles routing accuracy (mean DEG-weighted accuracy to ground truth) for the additive baseline versus the trained Stage-2 residual model, under the correlation metric (Pearson) and the magnitude-sensitive metric (CCC), split by subset. Stage-2 beats additive on the resistant subset under Pearson but not under CCC.

rebuild the source's approximately 8,192-gene panel from these files, so absolute per-metric BS off Norman is shifted upward. The metric is validated on Norman and faithful in structure, but exact cross-dataset agreement is out of reach with these inputs.

Reconstructed reliability filter. The reliability filter reimplements Appendix H, whose constants are not fully specified; it is flagged throughout as a reconstruction, and the biological-signal conclusion in Section 5 should be read with that caveat in mind.

One resistant double dataset. The gene-pair result (Section 4.7) rests on Wessels23. We attempted two independent replications and neither supplied a resistant class. Sunshine23's doubles proved too sparse for stable per-double pseudobulk (only 19 scorable, all saturated). The direct-capture replogle20 screen (GSE146194), processed under the identical protocol, is well-powered (37 gene-pair doubles, all with both singles measured, median 322 cells per double) but near-fully saturated: pooled median Baseline Saturation 0.989, with 0 of 37 doubles resistant and only 2 frontier, consistent with the source's near-ceiling additive weighted Pearson of 0.996 for this screen. Three of the four UnseenCombo screens examined here (Norman19, replogle20, Sunshine23) are thus saturated, and only Wessels23 exhibits a genuine resistant-doubles class. We read this as evidence of a field-level data scarcity in the resistant-doubles regime, not a weakness of the triage method: where a resistant class exists the triage predicts it (Wessels Spearman 0.50, resistant-versus-not AUC 0.77), but such screens remain rare in the public record.

Routing lift is small and metric-dependent. On the resistant Wessels doubles a trained residual Stage-2 gives a small positive routing lift under the correlation metric (oracle +0.007) but does not beat the additive baseline under the magnitude-sensitive metric, where the non-additivity is idiosyncratic per double. Replacing the residual model with GEARS, a purpose-built context-rich architecture, does not change this: GEARS matches the residual model's small lift under the Pearson metric and, like the residual model, falls short of the additive baseline under CCC (Section 4.6). We read this as evidence that a context-rich model does not help on a screen this small, not as evidence against context-rich models in general: with only 26 single perturbations and about 113 training doubles per fold,

GEARS is far below the genome-scale or 100-or-more-single screens it was designed for, and is data-starved here. Whether a context-rich Stage-2 raises the routing payoff on a larger resistant-doubles screen remains open; the triage that decides what to route is validated, and the ceiling on the routing payoff is set by Stage-2 quality and by the amount of resistant-doubles training data.

Regressor choice. Random Forest outperforms Ridge on the genome-scale Replogle K562 regression (0.650 vs 0.580) but not on the smaller sets: on Adamson and the Wessels gene-pair task Ridge matches or beats it, and the permutation test is run on Ridge. No single regressor is uniformly best, so rather than fix one we name the model at every headline and report the best-performing one per dataset, with Ridge as the conservative cross-check.

Inherited tier cutpoints. The 0.33 and 0.66 tier boundaries, and therefore the binarization of every classification task, are taken wholesale from Kalygina et al. without independent justification for our feature sets or datasets. A different cutpoint choice could shift the reported ROC-AUCs.

Aggregation and controls. We aggregate the seven metrics by the median and select the better of two positive controls per metric; both are documented choices that can shift tier boundaries.

7. CONCLUSION AND FUTURE WORK

We have demonstrated that Baseline Saturation, the per-perturbation diagnostic introduced by Kalygina et al., is predictable from pre-experimental, gene-intrinsic features. Our two-stage triage-and-routing framework provides a cheap, interpretable pre-screen for allocating experimental and modeling resources in single-cell perturbation studies. The genome-scale Replogle K562 result (Random Forest Spearman $\rho = 0.65$, ROC-AUC 0.86) and the Wessels gene-pair result (Spearman $\rho = 0.50$, resistant-versus-not AUC 0.77) establish that both single- and double-perturbation saturation can be forecast before the experiment is run.

Three directions extend this work. First, a context-rich Stage-2 model (GEARS, PRESAGE, or a larger trained network) that captures per-double magnitude non-additivity may increase the routing payoff on a larger resistant-doubles screen; on Wessels, GEARS did not (Section 4.6). Second, a second independent resistant-doubles screen would replicate the gene-pair headline beyond Wessels. Third, incorporating cell-type-specific features into the triage (rather than training per context) could improve cross-cell-type transfer beyond the current partial generalization.

Code and data availability: All results are produced from public datasets (GEARS-processed files on Harvard Dataverse, scPerturb on Zenodo, and GEO GSE146194) with fully documented scripts. The analysis code, methods notes, result tables, and figure scripts are available at `https://github.com/itzsam-lol/PGIP`. The processed datasets are not redistributed but are regenerated from public sources by the download and adapter scripts in the repository.

8. DECLARATION

Conflict of Interest: The author declares no conflict of interest.

Preprint status: This is a preprint. It has not been submitted to or peer reviewed by any journal or venue, and no such submission or acceptance is claimed or implied by this document.

Author Contribution: Satyam conceived the study, implemented the analysis pipeline, and wrote the manuscript.

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